Data Analysis
WS 2025

Electronics Engineering
Group : OHM SQUAD
Adarsh Anooj
Analram Valiyaparambil Sandeep

Semiconductor Sector Data Analysis

Motivation

- Rise of AI may have changed market structure.
- AI-focused firms differ in growth and risk from legacy semiconductor groups.
- How the semiconductor industry changed after the AI surge.
- Why this matters for investors and policymakers.
- How does this affect us ?

Dataset Overview

- **Dataset Name :** Semiconductor stocks and the AI Surge
- **Source :** Kaggle
- **Time Period :** January 2015 – August 2025 (Filtered from 2012 for relevance).
- **Sampling Frequency:** Daily (Trading Days).
- **Key Variables:** Close (Price), Volume (Liquidity), Stock name.
- **Observations:** 42606 rows (after filtering) & 9 columns.

Analyses Performed

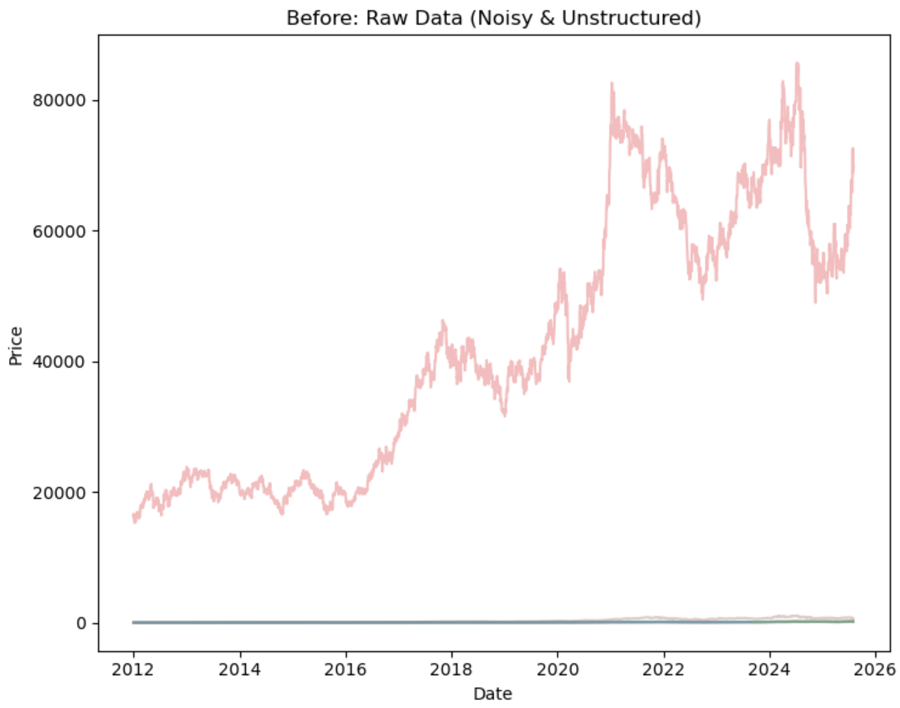
- Visualization and Exploratory Data Analysis (EDA)
- Probability and Event Analysis
- Statistical Theory Application
- Regression and Predictive Modelling
- Dimensionality Reduction
- Hypothesis Testing

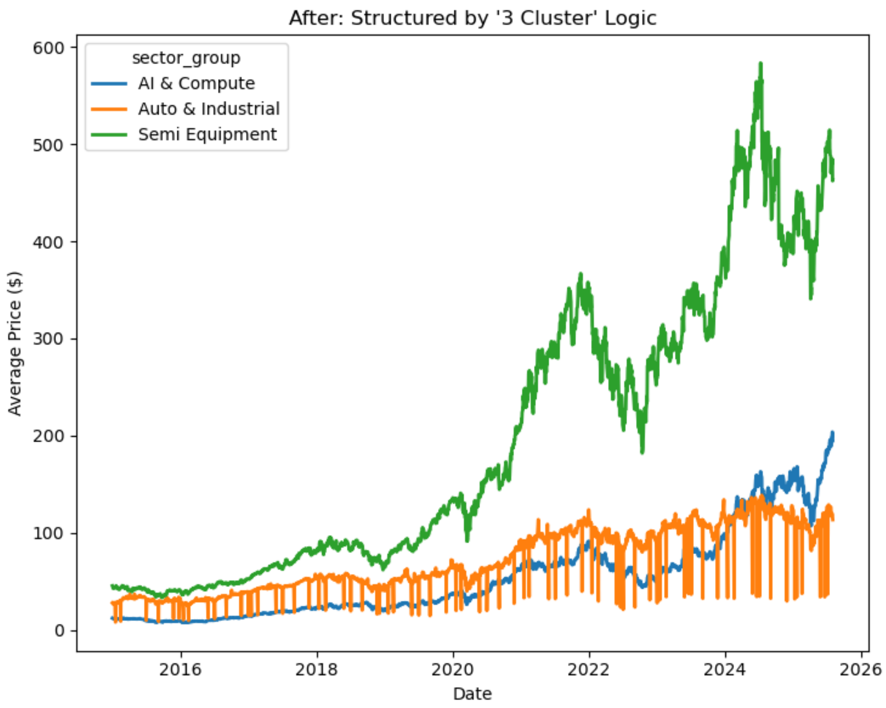
Data Quality & Preprocessing

- Found minor gaps during non-trading days and dropped them.
- Removed ghost columns.
- Focused on 2015+ to capture the modern GPU era.
- Segmented the industry into three : **AI & Compute** (e.g., NVDA), **Semi Equipment** (e.g., ASML), and **Auto & Industrial** (e.g., NXP).

FH | JOANNEUM
System Test Engineering

Before vs After Comparison





The Day that Left a Mark : AI Boom

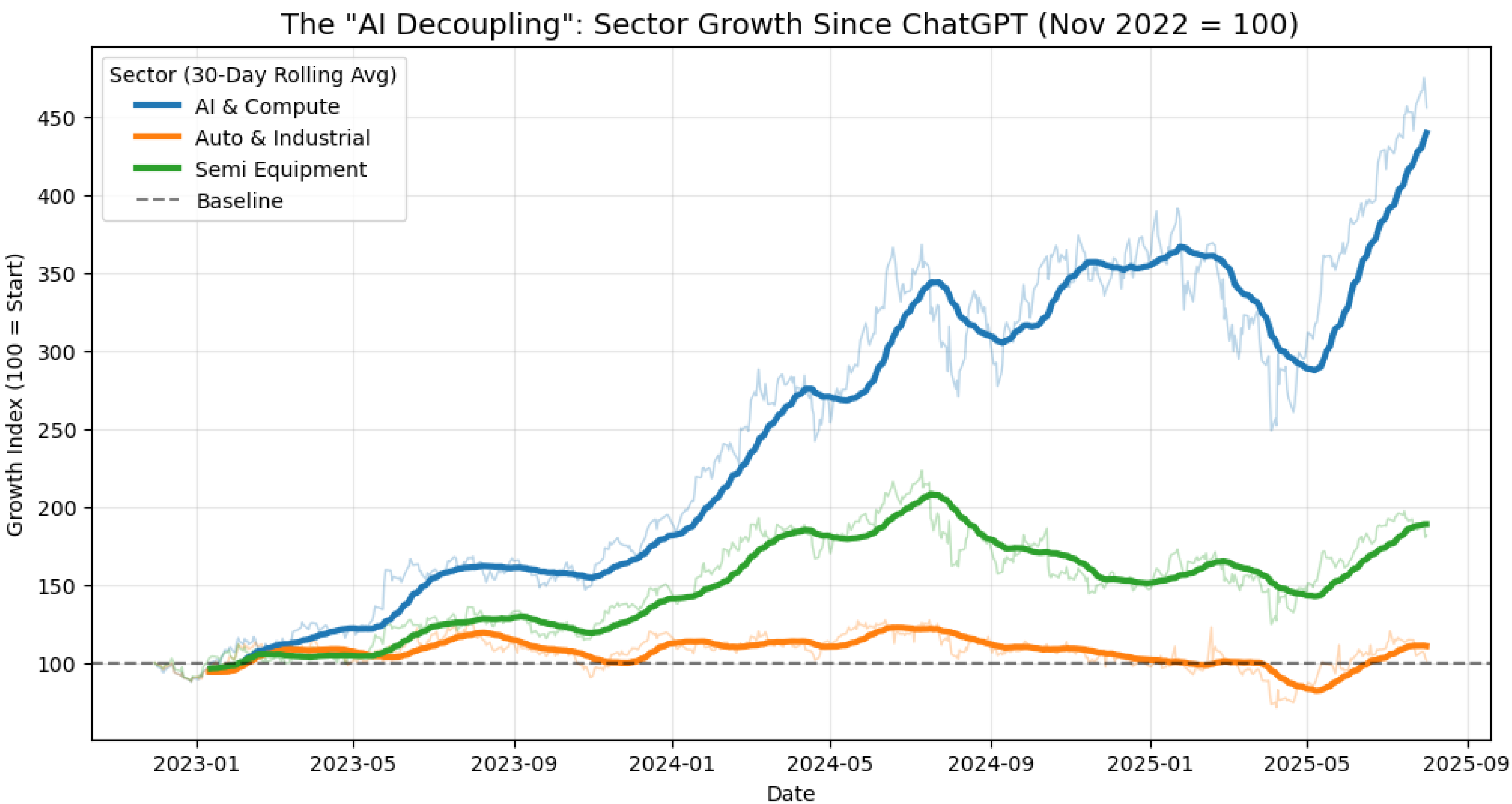
- ChatGPT officially released by OpenAI on **November 30, 2022.**
- Marked a visible inflection point in AI adoption and compute demand.
- Sudden increase in demand for GPUs and advanced accelerators.

Visualization and Exploratory Analysis

- Normalized prices to 100 (fair cross-company comparison) starting **Nov 30, 2022.**
- Evidence of a regime shift, not just a temporary spike.
- Main Observation:

-> AI & Compute : >250

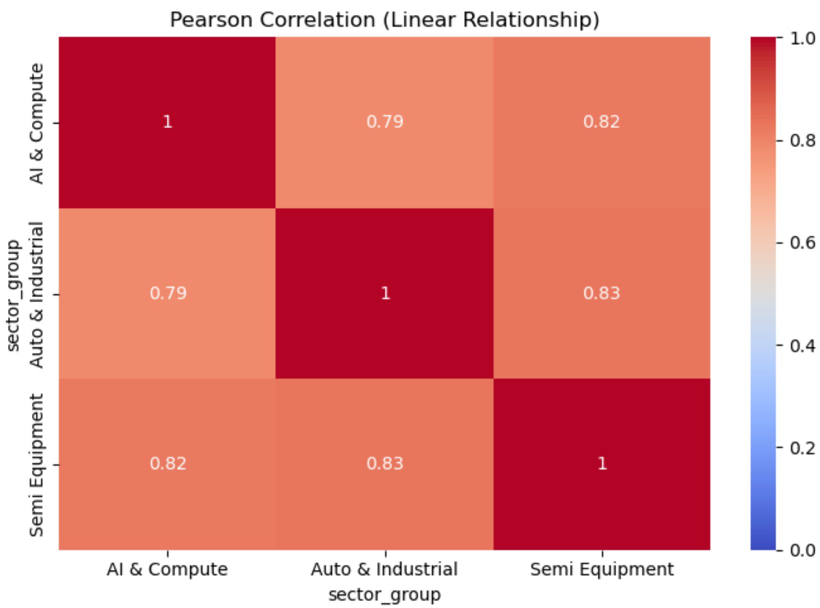
-> Auto & Industrial : ~110

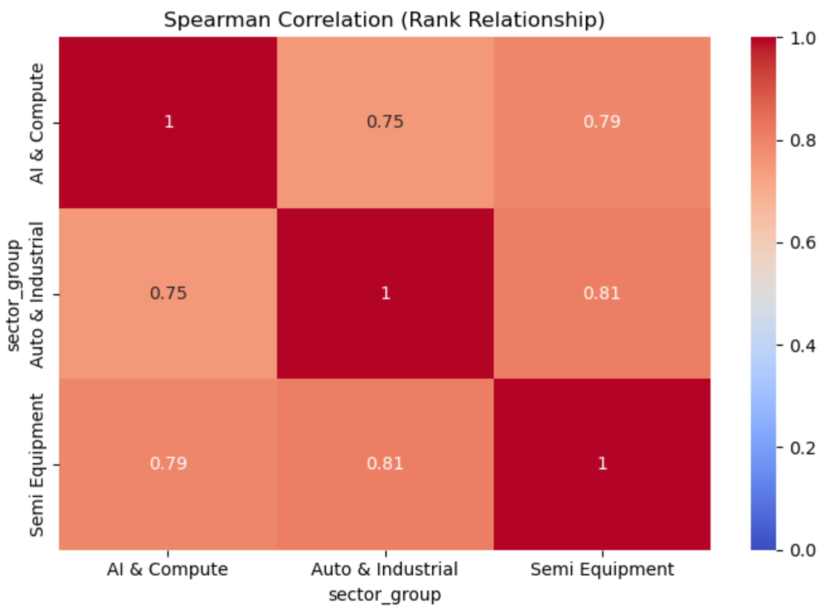


THE DIVERGENCE PLOT

Correlation and Standard Deviation

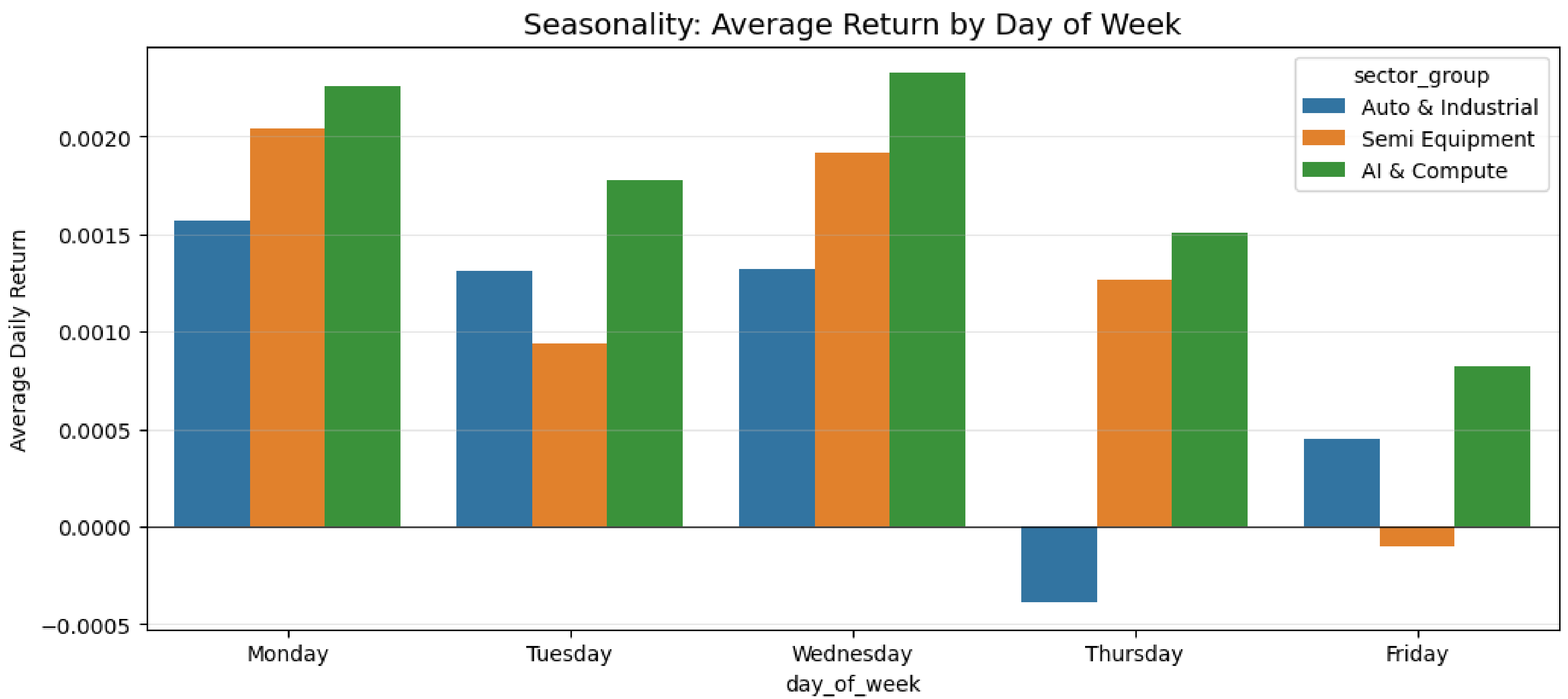
- Sectors are still highly correlated (>0.7) (Evidence: Pearson Correlation is ~0.78).
- AI Stocks are riskier than Auto Stocks (Evidence: AI Std Dev 0.0292 > Auto 0.0247)





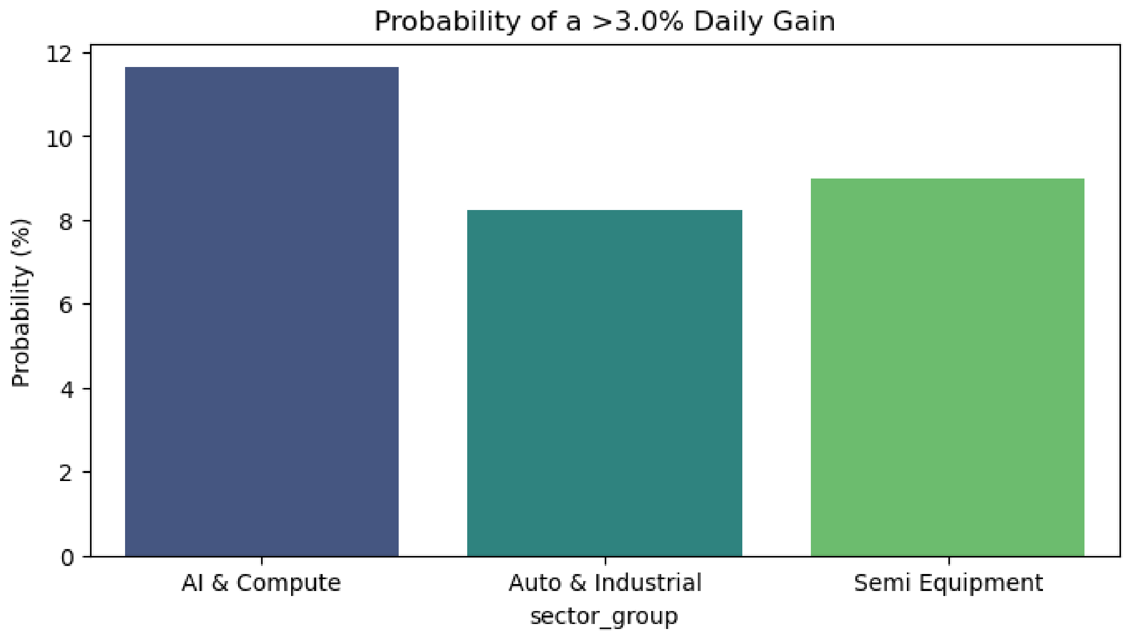
Seasonality & Trading Behavior

- Consistent **Friday** dip across all semiconductor sectors.
- Traders de-risk before weekends.



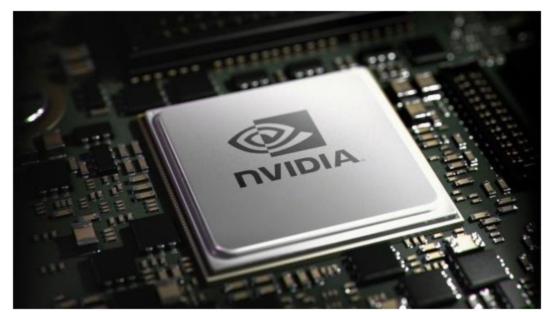
Probability and Event Analysis

- **AI** stocks have a significantly higher **'Super Day'** probability (~11%) vs Auto (~7%).
- Strong conditional link: If ASML is up, NVDA is ~75% likely to be up (Supply Chain effect).



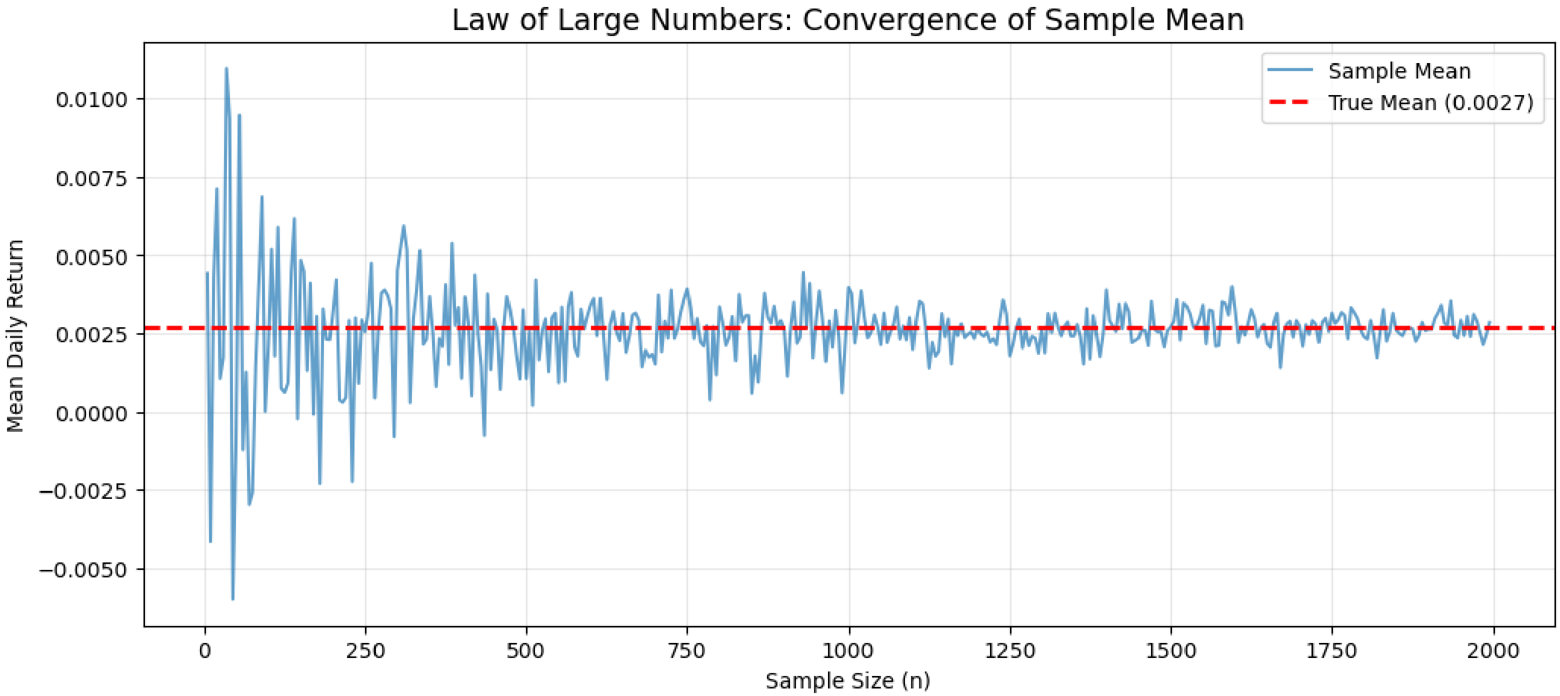
The Rise of NVIDIA

- Dominant supplier of AI training and inference GPUs.
- Direct beneficiary of the post-2022 AI infrastructure build-out.
- Connecting AI software demand & semiconductor supply.
- Used as the primary reference stock.
- Exhibits higher volatility and fatter tails than sector averages.



Law of Large Numbers

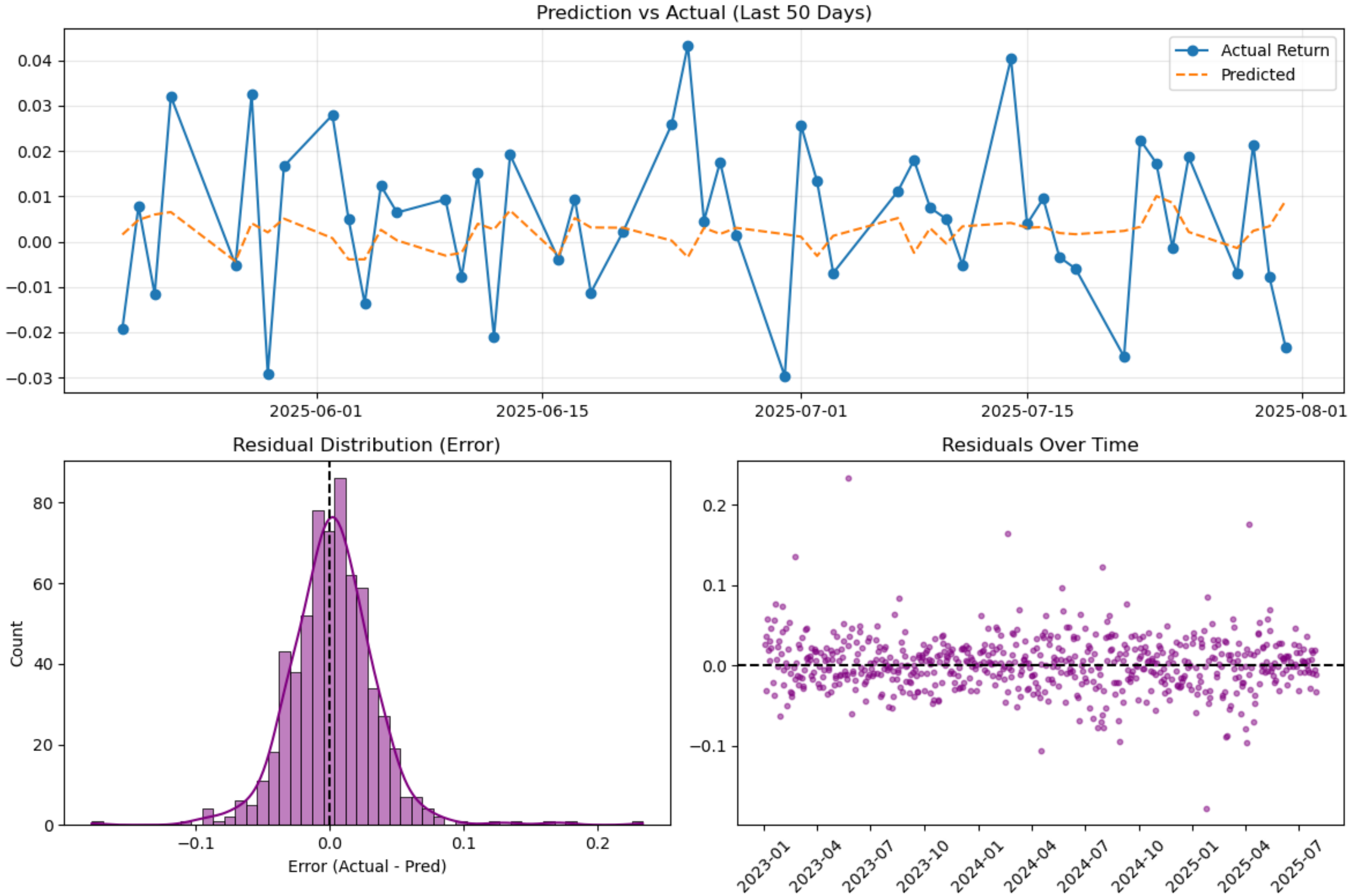
- At small sample sizes (n<100), the mean fluctuates wildly.
- As n increases, the sample mean converges to the True Mean.



Regression and Predictive Modelling

- Predict NVIDIA (NVDA) return.
- Used time-based train/test split (pre-2023 vs post-2023).
- We compared Linear Regression vs. Polynomial Regression.
- The model's failure is the finding.
- Large under-prediction during 2024 AI rally.
- Historical relationships failed post-AI boom.
- Model failure signals a regime change, not poor modeling.

Residual Analysis



Key Findings & Conclusions

- The semiconductor industry no longer behaves as a single sector.
- AI & Compute stocks have separated from Auto & Industrial.
- AI stocks exhibit higher growth and higher risk.
- The post-2022 AI boom represents a regime change.
- Models trained on pre-AI data fail to generalize to the new market structure.
- In short, AI did not just accelerate the semiconductor cycle : it rewrote the structure of the industry.